Project Initialization and Planning Phase

Date	25 October 2026
Team ID	EN25CS3010534
Project Title	Global Olympic Sports Performance Analysis
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build interactive Tableau dashboards that give sports strategists, investment analysts and journalists a clear, data-driven view of Olympic medals, participation by country and sport, regional efficiency, sports investment and host nations, so they can plan investments and tell stories with confidence.
Scope	Analysis of the four project datasets (athletes, medal results, Games summary and all-time country rankings) for 18 Olympic Games from 1992 to 2026, covering 60 countries, 50 sports, 7 regions, medals and host nations. Delivers two Tableau dashboards, a story and a Flask website for sports strategists, investment analysts and journalists. Out of scope: real-time data feeds and predictive models.
Problem Statement	
Description	Olympic data (athletes, countries, sports, medals and Games) is large and spread across many rows, files and years, so sports strategists, funding analysts and journalists struggle to see which countries, sports and regions perform best and where medals are won most efficiently.
Impact	Solving this gives decision-makers one reliable view of Olympic performance. It speeds up decisions, helps committees choose which sports to support, helps funders find regions with high medal returns, and gives journalists clear, data-backed stories.

Proposed Solution	
Approach	Connect Tableau to the four CSV files, check and prepare the data, create a Year field and a KPI Selector parameter, build a worksheet for each business question, combine them into two interactive dashboards with filters and tooltips, add a story, and publish to Tableau Public and a Flask website.
Key Features	KPI cards (athlete participation, sports categories entered, nations represented); world map of total medals by country; trend of total medals over the years; average medals per athlete by sport category; global performance by country with a Total, Gold, Silver and Bronze selector; regional KPIs; average medal efficiency by region; sports investment vs medal performance; medal totals across Olympic hosts; regional medal efficiency and distribution by sport; a Tableau story and a Flask website.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		

Computing Resources	CPU/GPU specifications, number of cores	Standard laptop or desktop with a multi-core processor (Intel Core i3 or higher); no GPU required
Memory	RAM specifications	4 GB minimum (8 GB recommended)
Storage	Disk space for data, models, and logs	At least 2 GB free disk space for the dataset, Tableau workbooks and the Flask project
Software		
Frameworks	Python frameworks	Tableau Desktop (Public Edition) for analysis and dashboards; Tableau Public for sharing; Flask for the website
Libraries	Additional libraries	Tableau built-in chart types (bar, line, map, pie, bubble, donut), calculated fields and parameters; Flask (Python); HTML and CSS
Development Environment	IDE, version control	VS Code for the Flask site; GitHub for version control; workbook saved as .twb / .twbx files
Data		
Data	Source, size, format	Four CSV files: athletes (8,000 rows, 26 fields), medal_results (910 rows, 19 fields), games_summary (18 rows, 10 fields) and alltime_country_rankings (60 rows, 13 fields); Games 1992-2026